Journal of Physics D: Applied Physics

The influence of interface state and energy barriers on the efficiency of heterojunction solar cells

To cite this article: H J Pauwels and G Vanhoutte 1978 *J. Phys. D: Appl. Phys.* **11** 649

J. Phys. D: Appl. Phys., Vol. 11, 1978. Printed in Great Britain. © 1978

The influence of interface states and energy barriers on the efficiency of heterojunction solar cells

H J Pauwels and G Vanhoutte
Laboratory of Electronics, University of Ghent, Ghent, Belgium

Received 21 June 1977, in final form 6 December 1977

Abstract. A heterojunction solar cell, in which interface recombination and arbitrary energy barriers at the interface occur, is analysed to find the optimum structure. Thermionic emission theory is used; it is assumed that one type of carriers are majority carriers at the interface and that their concentration can be known *a priori*. It is shown that these assumptions do not seriously limit the applicability of the theory. The results are discussed in the case when one semiconductor is much more heavily doped than the other, which is a condition that certainly improves the efficiency of the solar cell.

Diagrams are derived which allow one to determine the optimum combination of band-gaps for a given energy barrier at the interface, and the corresponding optimum efficiency. Two types of optimum structures are found. The first type behaves very much as a Schottky diode solar cell: all photons should be absorbed in the weakly doped semiconductor and its efficiency is maximum if this semiconductor is inverted at the interface. In the second type of optimum structure all photons should be absorbed in the heavily doped semiconductor, and its efficiency is maximum if the energy barrier at the interface is zero. For both types, these maximum efficiencies are lower than for optimum homojunction cells, because the diffusion velocity ($\approx 10^4$ cm s^{-1}), which appears in the saturation current of a homojunction, is replaced by either the thermal velocity ($\approx 10^7$ cm s^{-1}) or the interface recombination velocity (which maybe of the same order of magnitude).

1. Introduction

Various authors (e.g. Loferski 1956, Rappaport 1959, Wolf 1960) have investigated the efficiency of homojunction solar cells as a function of the band-gap. Until recently there has been no similar analysis for heterojunction solar cells. Thin-film solar cells do not exist as homojunction cells, so the heterojunction structure is a very important one. Since a heterojunction is characterised by two band-gaps, one can investigate which combination of band-gaps can in principle give rise to the highest efficiencies and how this combination must be incorporated in a cell structure such that its potential is fully used. Because of the large number of parameters influencing the efficiency of a solar cell, such an investigation is only possible if one manages by simplifying assumptions and approximations, to obtain analytical expressions and to classify the various parameters into groups that can independently be optimised. The assumptions should be such that they can in principle be realised in a solar cell structure, and that if they are not fulfilled, the efficiency is expected to be lower. The approximations should be such that they are expected to have minor influence on the results. It is not the purpose of such an investigation to analyse presently existing heterojunction structures, but to obtain optimum structures. The approximations must not therefore necessarily be justified for the existing structures, but only for optimum structures. It will be one of the results of the present article that the existing structures are not optimum.

Recently there has been a search for such optimum heterojunction structures (De Vos 1976, De Vos and Pauwels 1976, 1977). The essential assumptions were that no interface states existed at the junction, and that the energy barriers at the junction were such that they did not prevent collection of photon excited minority carriers. Since we shall need some of those results, we shall present here the main conclusions.

First, it was found that the separation of the efficiency into a spectrum factor and a voltage and curve factor, although very useful for homojunction cells, is misleading for heterojunction cells, because the advantage gained for a given structure in the spectrum factor is completely lost in the voltage and curve factor. Instead there was an analysis of the efficiency on the basis of the current–voltage characteristic given by

$$I = I_s\,[\exp(qV/kT) - 1] - I_L \tag{1}$$

where I_s is the saturation (particle) current (per unit area) and I_L the short-circuit light current. In their model, the saturation current is of the form

$$I_s = a_1 \exp(-E_{g1}/kT) + a_2 \exp(-E_{g2}/kT) \tag{2}$$

where E_{g1} and E_{g2} are the two band-gaps, and a_1 and a_2 the material parameters of each of the two semiconductors that are independent of the band-gaps. The inverse of the factor a could be called the 'quality' of the corresponding semiconductor. The short-circuit current is of the form

$$I_L = Q_1 r_1 I_1 + Q_2 r_2 I_2 \tag{3}$$

where I_1 and I_2 are the fluxes of photons that can be absorbed in each of the semiconductors for a given illumination, r_1 and r_2 the fractions that actually are absorbed, and Q_1 and Q_2 are the collection efficiencies. I_1 and I_2 depend on the band-gaps of the two semiconductors and on the choice of the illuminated side. Q_1, Q_2, r_1 and r_2 can to a great extent independently be controlled by the geometry of the device. On this basis, De Vos and Pauwels arrive at their second conclusion.

The material with the smaller band-gap should have the larger collection efficiency Q; it should absorb all photons ($r=1$) so that

$$I_L = QI(E_g) \tag{4a}$$

where E_g is the smaller band-gap and $I(E_g)$ the flux of photons in the incident light with energies larger than E_g; its quality should be the better (smaller a); its band-gap should be chosen optimally in the following way. This optimal choice depends on a and Q. In figure 1(a) this optimal choice of band-gap is shown as a function of a for $Q=1$. It is calculated by means of an analytical approximation of the AM1 spectrum (De Vos 1976). For $Q<1$, the optimal choice of band-gap corresponds to a/Q. The (optimal) efficiency that is obtained for this optimal choice of band-gap is shown in figure 1(b) as a function of a for $Q=1$. For $Q<1$, it is $\eta_Q = Q\eta\,(a/Q)$. The choice of the larger band-gap is not very critical (although its value may influence the required geometry of the cell). It should be sufficiently larger than the smaller band-gap such that the corresponding term in (2) is negligible, and thus

$$I_s = a \exp(-E_g/kT). \tag{4b}$$

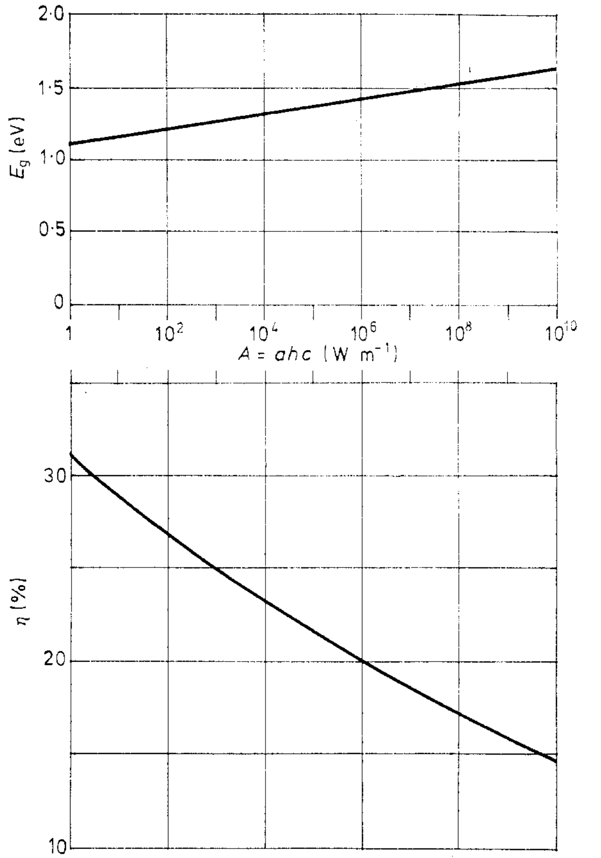

Figure 1. Optimal band-gap and optimal efficiency for a solar cell with I–V characteristic I_s [exp $(V)-1]-I_L$ where $I_s = a \exp(-E_g/kT)$ and I_L equals the flux of photons with energies larger than E_g of the incident light (AM1 spectrum).

This reveals the essential difference with a homojunction. For a homojunction

$$I_s = a_0 \exp(-E_g/kT) \qquad a_0 = a_1 + a_2 \tag{4c}$$

such that it is the semiconductor with the worse quality (larger a) which determines the optimal efficiency. For a good efficiency both semiconductors, at each side of the junction, should have a good quality. For the heterojunction, on the other hand, only the smaller band-gap material should have good quality, and the influence of a possible bad quality of the larger band-gap material can be eliminated by a sufficiently large choice of the band-gap.

The purpose of the present paper is to investigate to what extent these conclusions must be modified, because of the presence of interface states and arbitrary energy barriers at the junction of a heterojunction solar cell. Although it will be shown that interface states do deteriorate the efficiency, we believe that they are very difficult to avoid in purely thin film solar cells. The optimum structures that we shall derive will therefore be less good than those of De Vos *et al*, but they will be easier to realise. We nevertheless still make various simplifying assumptions and approximations (although much less than those used in the original articles on homojunctions: Loferski 1956, Rappaport 1959, Wolf 1960). We shall list them here, and mention them again when they are introduced in the model.

Generation by light of carriers in the space charge region is neglected. This may seem a serious limitation. It will be shown however, in a separate paper (Pauwels *et al* 1978) that incorporating this effect only requires an appropriate redefinition of some of the particle currents and does not change the conclusions.

The recombination of carriers in the space charge region is neglected. It is not verified whether this approximation is justified in all thin film structures, but incorporating this effect really excludes analytical treatment.

Majority carrier transport in the space charge region is treated with the thermionic emission theory instead of the thermionic emission–diffusion theory (see e.g. Sze 1969). The condition for this assumption is that the thermal velocity is much lower than the drift velocity at the junction. For thin film materials with their relatively low mobilities ($\mu \approx 100\,\mathrm{cm^2\,V\,s^{-1}}$), this condition might not be fulfilled: for instance for a band-bending of 1 eV, a doping concentration of $10^{17}\,\mathrm{cm^{-3}}$, a dielectric constant of 10 and a mobility of $100\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$, one easily finds for this drift velocity $10^7\,\mathrm{cm\,s^{-1}}$, which is of the same order of magnitude as the thermal velocity. We nevertheless accept this approximation because it greatly simplifies the equations; in a separate paper (Pauwels *et al* 1978) it will be shown that the thermionic emission–diffusion theory only requires appropriate redefinition of some of the thermal and recombination velocities used in the present theory.

At the junction, interface recombination from both conduction bands to the valence bands will be considered. Since the general expression for the interface recombination current is a nonlinear function of the carrier concentrations, the problem cannot be handled analytically. We therefore introduce an additional restriction: we assume that one of the semiconductors is much more heavily doped than the other. This assumption increases the efficiency, and is therefore not a restriction if one looks for optimum structures. Indeed, interface recombination currents increase the dark saturation current of the solar cell and decrease the collection efficiencies and thus the short-circuit current; they therefore decrease the efficiency of the solar cell; on the other hand it is well known from Shockley–Read theory that interface recombination currents are minimum if the interface is as extrinsic as possible. At the same time this assumption makes the problem analytically tractable: almost always it reduces the recombination current to the well known expression for minority carrier recombination, and makes the equilibrium concentration towards which the minority carriers relax an *a priori* known quantity; the few unrealistic situations in which this is not the case will be outlined in §4.

Tunnelling through energy spikes at the junction will not be incorporated into our model. Such tunnelling only occurs for spikes smaller than a few kT, and it can be made more or less important by adapting the doping of the semiconductors. In some models of the CdS–Cu_2S solar cell, this tunnelling plays an essential role in predicting the efficiency. We shall therefore relate our conclusions for such models to what really occurs (§5). In a separate paper (De Visschere and Pauwels 1978) a quantitative analysis of the influence of tunnelling will be presented, and will show that such tunnelling can indeed improve the efficiency of some models, but deteriorates the efficiency of what we shall find here to be optimum structures.

Under these assumptions the equations governing the model are derived in §2, and solved in §3. A few minor approximations of the obtained results are introduced in this section in order to stress the essential features of the results and to facilitate the discussion. These results are discussed in §4. In §5 our conclusions are applied to some experimental cells that are described in the literature. In §6 the main conclusions are summarised.

2. Model

We consider a p-type material forming a heterojunction with an n-type material (figure 2). The p-type material is on the left, the n-type material on the right; all quantities referring to the p-type material will have the subscript 1, those referring to the n-type material the subscript 2. The illumination of the device may be from the left or from the right. At the

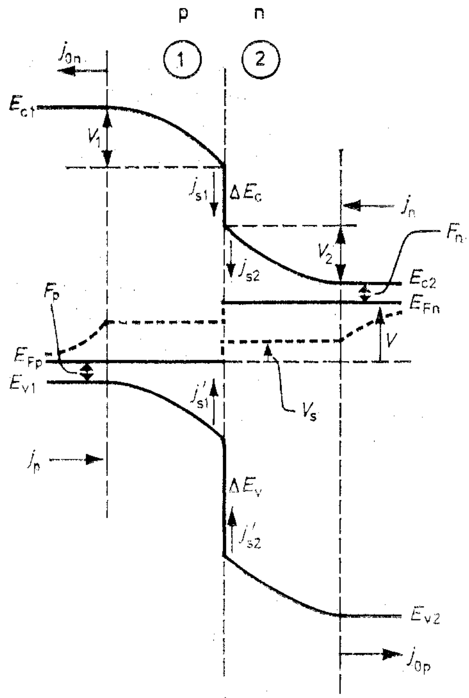

Figure 2. p–n heterojunction interface showing the bandbendings (V_1 and V_2), the discontinuities in the bands (ΔE_c and ΔE_v), the various particle current densities, the behaviour of the quasi-Fermi levels E_{Fn} and E_{Fp} under forward bias V, (the jump V_s will be neglected). F_n and F_p determine the doping concentrations.

junction there is a jump ΔE_c in the bottom of the conduction bands of the two semiconductors. This jump is determined by the difference in electron-affinities of both materials. It can be positive or negative; the ΔE_c shown in figure 2 is positive. We have chosen this sign convention because such a positive ΔE_c constitutes no barrier to the collection by the n-type material of light generated electrons in the p-type material. Likewise there is a jump ΔE_v in the top of the valence bands in the two semiconductors. This jump $\Delta E_v = \Delta E_c + \Delta E_g$ where $\Delta E_g = E_{g2} - E_{g1}$ is the difference in the band-gaps of the two semiconductors. Again, this ΔE_v can be positive or negative. The ΔE_v shown in figure 2 is positive; such a positive ΔE_v constitutes no barrier to the collection by the p-type material of light-generated holes in the n-type material. The band-bending over the amounts V_1 and V_2 in the space charge regions of the p-type and the n-type material respectively, is determined by the charge neutrality condition for the charges in the two space charge regions and in interfaces states and possible inversion layers at the junction. These V_1 and V_2 depend, in a more or less complicated way on the applied voltage V over the p–n junction. In figure 2 a forward applied voltage V is shown. For simplicity of notation, all energies and voltages will be normalised to kT and kT/q (=25 mV) respectively.

The thermal equilibrium majority and minority carrier concentrations of the semiconductors 1 and 2 will be called respectively p_1, n_1 and n_2, p_2.

Since recombination and generation is neglected in the space charge regions, one has

$$j_n - j_{s2} = j_{s1} + j_{0n} \tag{5}$$

$$j_p - j_{s1}' = j_{s2}' + j_{0p} \tag{6}$$

where j_n, j_{s1}, j_{s2} and j_{0n} are electron particle current densities respectively entering the space charge region of the n-type material, recombining from the conduction bands 1 and 2 at the interface, and leaving the space charge region of the p-type material. The quantities j_p, j_{s1}', j_{s2}' and j_{0p} are hole particle current densities defined analogously with naturally

$$j_{s1} + j_{s2} = j_{s1}' + j_{s2}' = j_s. \tag{7}$$

Since thermionic emission theory is accepted, the electron and hole quasi-Fermi levels are flat in the space charge regions, have a drop at the interface, and have a final drop in the neutral regions of the p- and n-type materials where minority carrier transport occurs through diffusion. From these assumptions about the behaviour of the quasi-Fermi levels it follows that from the four carrier concentrations at the interface, n_{1s}, n_{2s}, p_{1s} and p_{2s} only two are unknown, i.e. n_{1s} and p_{2s}. The other two carrier concentrations are indeed given by

$$n_{2s} = n_2 \exp(-V_2) \tag{8}$$

$$p_{1s} = p_1 \exp(-V_1). \tag{9}$$

The two continuity equations (5) and (6) can be used to determine these unknowns. All the currents in the equations can indeed be expressed in terms of these unknowns, but unfortunately the expressions for the recombination currents are nonlinear. Only if two conditions are both fulfilled, the equations (5) and (6) become linear in the unknowns: (1) if one type of carriers, say the holes, are majority carriers at the interface, i.e.

$$p_{1s} + p_{2s} \gg n_{1s},\ n_{2s} \tag{10a}$$

and (2) if the drop in the corresponding quasi Fermi-level at the interface, thus in E_{Fp}, is negligible. Only *a posteriori* (see Appendix) we shall be able to show that this second condition requires, if

$$\Delta E_c > 0: p_1 \exp(-V_1)\, v_{t1} \gg n_2 \exp(-V_2)\, s_2 \qquad V_2 + \Delta E_v > 0 \tag{10b}$$

or if

$$\Delta E_c < 0: p_1 \exp(-V_1)\, v_{t1} \gg n_2 \exp(-V_2)\, v_{t2} \qquad V_2 + \Delta E_v > 0 \tag{10c}$$

where v_{t1} and v_{t2} are the thermal velocities $[(kT/2\pi m^*)^{1/2}]$ of holes and electrons in semiconductors 1 and 2 respectively, and where s_2 is the interface recombination velocity from the conduction band 2. The conditions (10) are likely to be fulfilled if the p-type semiconductor is much more heavily doped than the n-type semiconductor so that $V_1 \ll V_2$. As explained in the introduction, condition (10*a*) minimises the interface recombination currents and therefore improves the solar cell efficiency. Notice that neglecting the drop in E_{Fp} at the interface also implies that p_{2s} is no longer unknown:

$$p_{2s} = p_2 \exp(V_2 + V). \tag{11}$$

The continuity equation (6) is thus no longer needed. Under these conditions (10), the Shockley–Read theory leads to

$$j_{s1} = s_1(n_{1s} - n_{1s0}) = s_1[n_{1s} - n_1 \exp(V_1)] \tag{12}$$

$$j_{s2} = s_2(n_{2s} - n_{2s0}) = s_2[n_2 \exp(-V_2) - n_2 \exp(-V_2 - V)] \tag{13}$$

where n_{1s0} and n_{s20} are the electron densities at the interface calculated with the (single) hole quasi-Fermi level, and where s_1 and s_2 are the interface recombination velocities from the conduction bands 1 and 2 respectively. These s_1 and s_2 are proportional to the density of active traps at the interface; that is, the traps that lie in the range $E_{gs} - 2F_{ps}$ above E_{Fp} (where E_{gs} is the energy difference between the lower conduction band and the higher valence band at the interface and F_{ps} the energy difference between E_{Fp} and this higher valence band). We shall show later that V_1 is independent of V, so that also this energy range and thus s_1 and s_2 are independent of V.

Thermionic emission theory leads to the following expression for the electron particle current entering semiconductor 1:

$$j_n - j_{s2} = [n_2 \exp(-V_2) - n_{2s}']v_t \exp(-\Delta E_c), \qquad \Delta E_c > 0 \tag{14a}$$

$$= [n_2 \exp(-V_2) - n_{2s}']v_t, \qquad \Delta E_c < 0 \tag{14b}$$

where

$$v_t = v_{t2} - s_2 \tag{14c}$$

and where n_{2s}' is the electron concentration at the interface of semiconductor 2 calculated with the electron quasi-Fermi-level of semiconductor 1. Therefore

$$n_{1s} = \beta_c \exp(-\Delta E_c) n_{2s}', \qquad \beta_c = N_{c1}/N_{c2} \tag{15}$$

where N_{c1} and N_{c2} are the equivalent densities of states of the conduction bands of semiconductor 1 and 2 respectively. n_{2s}' is thus not a new unknown.

Finally, the diffusion equation for minority carriers in the neutral regions in the presence of illumination, leads to

$$j_{0n} = [n_{1s} \exp(-V_1) - n_1]v_{d1} - \gamma_1 r_1 I_1 \tag{16}$$

$$j_{0p} = p_2 v_{d2} [\exp(V) - 1] - \gamma_2 r_2 I_2. \tag{17}$$

The quantities r_1, I_1, r_2 and I_2 have been defined in equation (3); γ_1 and γ_2 are the collection efficiencies by the space charge region of the photon excited minority carriers; they differ from Q_1 and Q_2 of equation (3) by the fact that not all carriers that are collected by the space charge region reach the other semiconductor; v_{d1} and v_{d2} are the diffusion velocities in the two semiconductors. Expressions for γ and v_d can be found in the literature (Te Velde 1973). They depend on the mobility μ, the minority carrier lifetime τ, the surface recombination velocity at the outer surface, the absorption coefficient and the thickness of the semiconductor but not on the band-gap. Depending on the thickness, v_d varies between the surface recombination velocity and $D/L = (\mu\, kT/q\tau)^{1/2}$. For $\mu = 40\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$, $\tau = 10^{-8}$ s, one has $D/L = 10^4\ \text{cm s}^{-1}$, i.e., three orders of magnitude smaller than $v_t \approx 10^7\ \text{cm s}^{-1}$.

3. Results

All current expressions of equation (5) have now been given in terms of one unknown n_{1s} (or n_{2s}'), i.e. by the equations (12), (13), (14) and (16). Solving this equation for n_{1s} and substituting the solution into the expression

$$I = j_n + j_{op} \tag{18}$$

gives the I–V characteristic of the solar cell. Somewhat simpler expressions for the result can be obtained if one takes into account the relation between the equilibrium concentrations n_1 and n_2:

$$n_1 \exp(V_1) = \beta_c n_2 \exp(-V_2 - V - \Delta E_c), \qquad \beta_c = N_{c1}/N_{c2}. \tag{19}$$

One obtains

$$I = [\exp(V) - 1][p_2 v_{d2} + s_2 n_2 \exp(-V_2 - V) + \delta(s_1 n_1 \exp(V_1) + n_1 v_{d1})] - \gamma_2 r_2 I_2 - \delta \gamma_1 r_1 I_1 \tag{20}$$

where

$$\delta = \frac{v_t}{v_t + s_1\beta_c + \exp(-V_1) v_{d1} \beta_c} \qquad \Delta E_c \geqslant 0 \tag{21a}$$

$$= \frac{v_t \exp(\Delta E_c)}{v_t \exp(\Delta E_c) + s_1 \beta_c + \exp(-V_1) v_{d1} \beta_c} \qquad \Delta E_c \leqslant 0. \tag{21b}$$

The quantity δ can be interpreted as a transport efficiency through the interface for electrons excited in the p-type region. In the Appendix we shall also need the result for the total recombination current $j_s = j_{s1} + j_{s2}$. One obtains

$$j_s = n_2 \exp(-V_2 - V)\,[\exp(V) - 1][s_2 + \beta_c \exp(-\Delta E_c)\delta s_1] + \delta_s \gamma_1 r_1 I_1 \tag{22}$$

where δ_s has the same denominator as δ but $s_1\beta_c$ as numerator. The essential features of this result for I can be stressed by making the following assumptions:

If $\Delta E_c > 0$ we shall suppose that

$$v_t \gg s_1 \beta_c + \exp(-V_1) v_{d1} \beta_c \tag{23a}$$

so that

$$\delta \approx 1. \tag{23b}$$

We have indeed mentioned in §2 that v_t is three orders of magnitude larger than v_{d1}. The interface recombination velocity s_1 can be of the same order of magnitude as v_t, but we shall suppose that it is, e.g. one order of magnitude smaller than v_t. One then obtains from equation (20)

$$I = [\exp(V) - 1][p_2 v_{d2} + s_c n_2 \exp(-V_2 - V) + n_1 v_{d1}] - \gamma_1 r_1 I_1 - \gamma_2 r_2 I_2 \tag{24}$$

where s_c is an effective recombination velocity defined for $\Delta E_c > 0$ by

$$s_c = s_2 + s_1 \exp(-\Delta E_c)\beta_c \approx s_2. \tag{25}$$

If, on the other hand, $\Delta E_c < 0$, we shall suppose that

$$v_t \exp(\Delta E_c) \ll s_1 \beta_c + v_{d1} \exp(-V_1) \beta_c \tag{26}$$

so that

$$\delta \approx 0. \tag{27}$$

It is clear that the transition from condition (23a) to (26) requires that ΔE_c drops at least a few kT below zero. One then obtains from equation (20)

$$I = [\exp(V) - 1][p_2 v_{d2} + v_{t2} n_2 \exp(-V_2 - V)] - \gamma_2 r_2 I_2. \tag{28}$$

If however for very small s_1 or very small negative ΔE_c the inequality sign in equation (26) is reversed, so that $\delta \approx 1$, one obtains

$$I = [\exp(V) - 1][p_2 v_{d2} + s_c n_1 \exp(V_1) + n_1 v_{d1}] - \gamma_1 r_1 I_1 - \gamma_2 r_2 I_2 \tag{29a}$$

where now s_c is defined for $\Delta E_c < 0$ by

$$s_c = s_1 + s_2 \exp(\Delta E_c)/\beta_c. \tag{29b}$$

4. Discussion

For the case $\Delta E_c > 0$, the discussion will be based on equation (24), which is valid under the conditions (10) and (23a). For the case $\Delta E_c < 0$, the discussion will be based on equation (28), which is valid under the conditions (10) and (26).

From equations (24) and (28) it follows that the I–V characteristic is of the form of equation (1) only if V_1, and thus according to equation (19) also $V_2 + V$, is independent of V. Fortunately this condition is compatible with the assumption of high relative doping of the p-type material. Indeed, if no inversion layer occurs in the n-type material, the band-bending in the p-type material is negligible, i.e. $V_1 = 0$, because the charge density in the depletion layer of the p-type material is much larger than that of the n-type material. If however an inversion layer is present in the n-type material, its charge density may become comparable with the charge density in the depletion layer of the p-type material, and it will no longer be true that $V_1 = 0$. But because charge density in an inversion layer increases exponentially with band-bending, we shall make the approximation that the top of the valence band of the n-type material stays fixed on the value at which its hole concentration equals the doping concentration of the p-type material. This situation is shown in figure 3. In this case $V_1 = -\Delta E_v = -\Delta E_g - \Delta E_c$ and is again independent of V. We conclude that

$$V_1 = 0 \quad \text{if} \quad \Delta E_v = \Delta E_g + \Delta E_c > 0 \tag{30a}$$

$$V_1 = -\Delta E_g - \Delta E_c \quad \text{if} \quad \Delta E_v = \Delta E_g + \Delta E_c < 0 \tag{30b}$$

and that the I–V characteristic is of the form of equation (1). We investigate now the form of I_s and I_L.

If $\Delta E_c > 0$, then, according to equation (24),

$$I_L = \gamma_1 r_1 I_1 + \gamma_2 r_2 I_2. \tag{31}$$

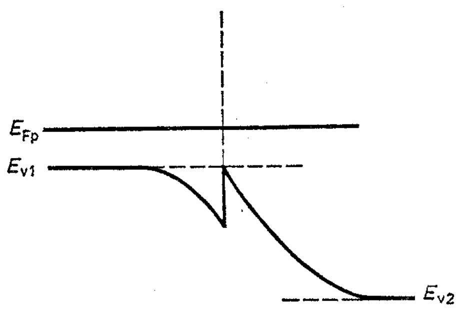

Figure 3. Energy diagram of the valence band if an inversion layer is present in semiconductor 2.

If one assumes that by appropriate choice of the geometry of the solar cell, one can, independently of each other, vary r_1 and r_2 between 0 and 1, and make a choice for γ_1 and γ_2 between a larger and smaller value, it is easily demonstrated that for a given choice of band-gaps, I_L is maximum if the smaller band-gap material absorbs all photons and has the larger γ. This maximum value equals

$$I_L = \gamma I(E_g) \tag{32}$$

where E_g is the smaller band-gap, $I(E_g)$ the flux of photons with energies larger than E_g incident on the solar cell, and γ the collection efficiency of the smaller band-gap material. Because this I_L depends only on the band-gap of the smaller band-gap material, we express I_s as

$$I_s = a \exp(-E_g). \tag{33}$$

One must consider three situations. If $\Delta E_g = E_{g2} - E_{g1} > 0$, then $E_g = E_{g1}$, and also $\Delta E_v = \Delta E_c + \Delta E_g > 0$ so that $V_1 = 0$. Therefore it follows from equation (24) that

$$a = N_{c2} \exp(F_p)\, s_c \exp(\Delta E_c) + N_{c1} \exp(F_p)\, v_{d1} + N_{v2} \exp(F_n)\, v_{d2} \exp(-\Delta E_g). \tag{34a}$$

If $\Delta E_g < 0$, then $E_g = E_{g2}$. If $\Delta E_v = \Delta E_c + \Delta E_g > 0$ so that $V_1 = 0$, it follows from equation (24) that

$$a = N_{c2} \exp(F_p) s_c \exp(\Delta E_c + \Delta E_g) + N_{c1} \exp(F_p) v_{d1} \exp(\Delta E_g) + N_{v2} \exp(F_n) v_{d2}. \tag{34b}$$

Finally if $\Delta E_v = \Delta E_c + \Delta E_g < 0$, inversion occurs, $V_1 = -\Delta E_c - \Delta E_g$, and it follows from equation (24) that

$$a = N_{c2} \exp(F_p) s_c + N_{c1} \exp(F_p) v_{d1} \exp(\Delta E_g) + N_{v2} \exp(F_n) v_{d2}. \tag{34c}$$

These three cases are presented on figure 4. Only the dominant term of a is indicated.

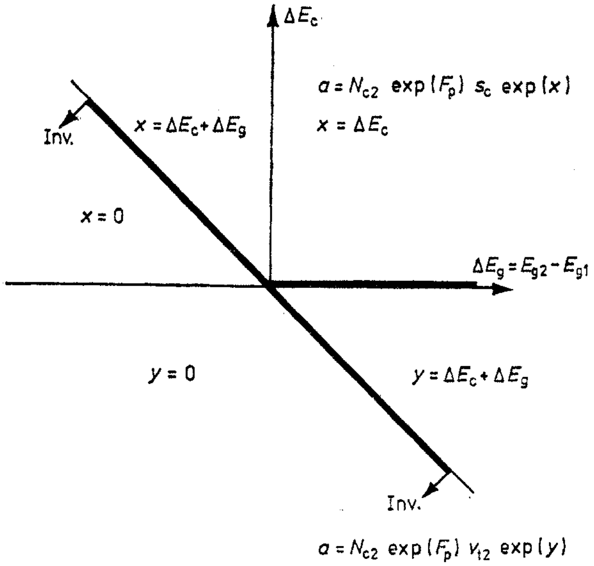

Figure 4. ΔE_c, ΔE_g diagram for a p$^+$n junction. ΔE_v is measured by the vertical distance to the bisector which shows the onset of inversion. The dominant term of the a-factor defined in figure 1 is shown in the various sectors of the diagram. The optimal band-gap E_g of figure 1 is the smaller band-gap for $\Delta E_c > 0$ and the band-gap E_{g2} for $\Delta E_c < 0$. The heavy lines show the structures with the optimal values of the a-factor ($X = 0$ or $Y = 0$).

If $\Delta E_c < 0$, then according to equation (28)

$$I_L = r_2 \gamma_2 I_2 \tag{35}$$

which by appropriate choice of geometry, can be made maximum if the n-type material absorbs all photons and has the larger γ. This maximum value is then

$$I_L = \gamma_2 I(E_{g2}). \tag{36}$$

Because this I_L depends only on E_{g2}, we express I_s as

$$I_s = a \exp(-E_{g2}). \tag{37}$$

One must consider two cases. If $\Delta E_v = \Delta E_g + \Delta E_c > 0$, then $V_1 = 0$ and from equations (28) and (19), it follows that

$$a = N_{c2} \exp(F_p)\, v_{t2} \exp(\Delta E_g + \Delta E_c) + N_{v2} \exp(F_n)\, v_{d2}. \tag{38a}$$

If on the other hand, $\Delta E_v = \Delta E_g + \Delta E_c < 0$, then inversion occurs, $V_1 = -\Delta E_g - \Delta E_c$ and from equations (28) and (19) it follows that

$$a = N_{c2} \exp(F_p)\, v_{t2} + N_{v2} \exp(F_n)\, v_{d2}. \tag{38b}$$

Also these two cases are presented on figure 4. Only the dominant term of a is indicated.

For a given E_{g1}, E_{g2} and ΔE_c one can find the corresponding value of a on figure 4 or in equations (34) and (38). In figure 5 one then finds the efficiency of the corresponding solar cell with optimum geometry, i.e. for which I_L has the form (32) or (36). The curves in figure 5 are plotted for various values of a and in function of either the smaller band-gap (if $\Delta E_c > 0$) or the band-gap E_{g2} (if $\Delta E_c < 0$). They are valid for the corresponding γ equal to 1. If $\gamma < 1$ one finds

$$\eta_\gamma = \gamma \eta\, (a/\gamma). \tag{39}$$

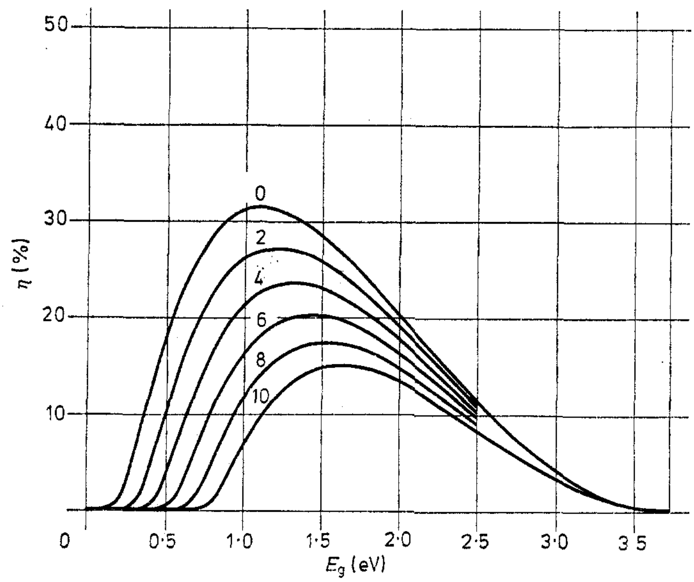

Figure 5. Efficiency of a solar cell with I–V characteristic as defined in figure 1 in function of the band-gap E_g and the a-factor. Curve parameters give values of lg A where A (W m^{-1})$=ahc$. $T = 293$ K.

For a given ΔE_g and ΔE_c, one can find from figure 1(*a*) the smaller band-gap (if $\Delta E_c > 0$) or the band-gap E_{g2} (if $\Delta E_c < 0$) that gives rise to an optimal efficiency, and from figure 1(*b*) this corresponding optimal efficiency for $\gamma = 1$. If $\gamma < 1$ they correspond to a/γ and equation (39).

Furthermore, for a given ΔE_c one can find the corresponding range of ΔE_g that gives rise to the best efficiencies, or vice versa, for a given ΔE_g one can find the corresponding range of ΔE_c that gives rise to the best efficiencies. The best efficiencies correspond with the lowest value of a. It is an important conclusion that these lowest values of a are obtained in two types of situations: (1) if the weakly doped semiconductor is inverted at the interface, and (2) for positive ΔE_g, if ΔE_c approaches zero from the positive side.

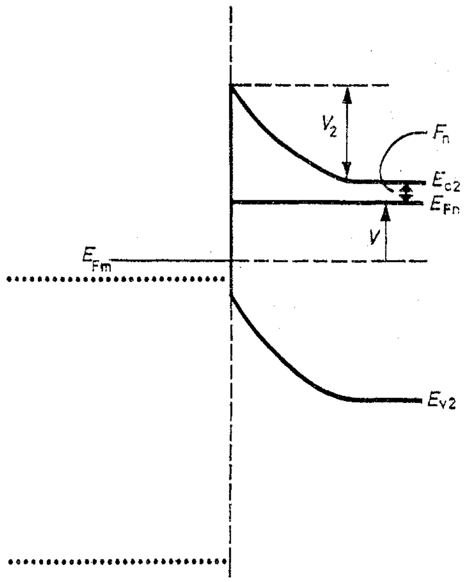

Figure 6. Energy band diagram of a Schottky diode formed by a metal and an n-type semiconductor under forward bias V. The dotted lines show the conduction and valence band edges for the oxides forming isotype heterojunctions with the n-type semiconductor as described at the end of section 5.

For the first type, all photons should be absorbed in the weakly doped semiconductor and its band-gap should be optimal (i.e. correspond to the value of a). For the second type, all photons should be absorbed in the heavily doped semiconductor and its band-gap should be optimal. From equations (24) and (28), it follows that the dominant term of a decreases exponentially with $V_2 + F_n + V$, i.e. with the energy difference at the interface between the conduction band of the n-type material and the hole Fermi level of the p-type material. For the first type of structure, with the band-gap of the n-type material optimal, this energy difference is maximum if the n-type material is inverted. For the second type of structures, with the band-gap of the p-type material optimal and $\Delta E_c \geqslant 0$ (such that $Q_1 = \gamma_1$), this energy difference is maximum if $\Delta E_c = 0$. One should notice the great resemblance between Schottky diodes and the present heterojunctions. The dark current of a Schottky diode is (figure 6)

$$I = N_{c2} v_{t2} \exp(-V - F_n - V_2)[\exp(V) - 1] \tag{40a}$$

while the dark current of the heterojunction, if one takes only the dominant term of a

into account, is given by

$$I = N_{c2} s_c \exp(-V - F_n - V_2)[\exp(V) - 1] \qquad \text{for } \Delta E_c > 0 \tag{40b}$$

and

$$I = N_{c2} v_{t2} \exp(-V - F_n - V_2)[\exp(V) - 1], \qquad \text{for } \Delta E_c < 0 \tag{40c}$$

the difference being that for a Schottky diode $V + F_n + V_2$ is determined by the difference in the work-function of the metal and the electron affinity of the semiconductor, while for the heterojunction $V + F_n + V_2$ is also determined by the band-bending in the p-type semiconductor.

The first type of structures can be subdivided in those with $\Delta E_c > 0$ and those with $\Delta E_c < 0$. For the latter type the analogy with the Schottky diode solar cell is almost complete because the photons must be absorbed in the weakly doped semiconductor and the dark saturation current is determined by v_{t2}. We shall later refer to these latter type structures as MS-type structures.

A few final remarks are in order. The conclusions in the papers of De Vos, mentioned in the introduction, are valid for the case $\Delta E_c \geqslant 0$ and $s_c = 0$. It can then easily be seen from equations (34) that the major contribution to a comes indeed from the smaller band gap material, so that the 'quality' of the larger band-gap material becomes unimportant. A normal value of $A = ahc$ for this case is 10^4 W m^{-1}. By introducing s_c 100 times larger than v_d, the best value of A (i.e. for inversion) becomes 10^6 W m^{-1}; that is, we go from curve 4 in figure 5 to curve 6. The case $\Delta E_c < 0$ and $s_c = 0$ has not been discussed so far because of condition (26). If the inequality sign in (26) is reversed, i.e. if s_1 is extremely small and if $V_1 + \Delta E_c \geqslant 0$, equation (29) is valid. The conclusions of De Vos *et al* for $s_c = 0$ therefore remain valid even if ΔE_c and ΔE_v are negative, as long as $V_1 + \Delta E_c \geqslant 0$ and $V_2 + \Delta E_v \geqslant 0$. If there is no interface recombination, there is no reason to dope one semiconductor more heavily than the other, so that one can try to adjust the dopings so that $V_1 + \Delta E_c \geqslant 0$ and $V_2 + \Delta E_v \geqslant 0$ are both fulfilled.

The present theory is valid under certain assumptions, all related to the fact that the p-type semiconductor is much more heavily doped than the n-type semiconductor. These assumptions are stated in equations (10). These limits of applicability of our theory are indicated in the $(\Delta E_c, \Delta E_g)$ plane presented in figure 7. We have assumed that

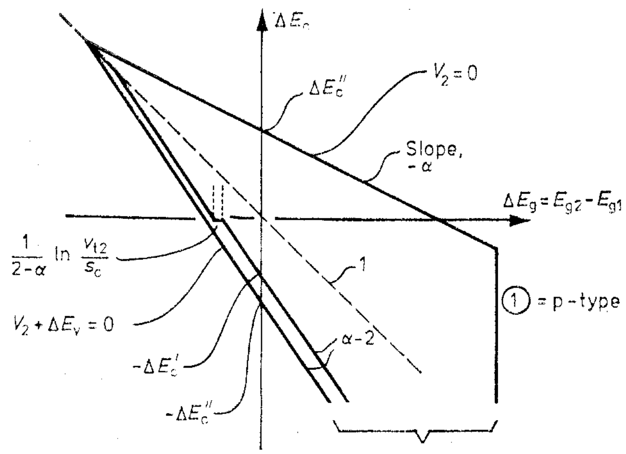

Figure 7. Limits of applicability of the theory in the $\Delta E_c - \Delta E_g$ diagram as described by the equations (41).

$E_{g1}=E_g-\alpha\Delta E_g$, $E_{g2}=E_g+(1-\alpha)\Delta E_g$, with E_g constant. The intersections of these limits with the ΔE_c axis are respectively

$$\Delta E_c'=E_g-2F_p-V+\ln(N_{v1}v_{t1}/N_{c2}v_{t2}) \tag{41a}$$

$$\Delta E_c''=E_g-F_p-F_n-V. \tag{41b}$$

The straight lines crossing the ΔE_c axis at $\pm\Delta E_c''$ express respectively $V_2 \geqslant 0$ and $V_2+\Delta E_v \geqslant 0$ and are limits in most cases. Only when $\Delta E_c' < \Delta E_c''$ the broken line crossing the ΔE_c-axis at $-\Delta E_c'$ serves as a limit; this occurs only when $p_1 v_{t1} < s_c n_2$ for $\Delta E_c > 0$, and when $p_1 v_{t1} < n_2 v_{t2}$ for $E_c < 0$; this limit expresses the first of the conditions (10*b*) or (10*c*). The vertical line at the right is described by

$$\alpha\Delta E_g=E_g-2F_p+\ln(N_{c1}/N_{v1}) \tag{41c}$$

and expresses that semiconductor 1 is p-type. All other conditions are automatically fulfilled. All these statements can easily be proved by elementary calculations. Most of these limits depend on V. Since our theory need only be valid up to the maximum power point, one may take V to be about 50% of the relevant band-gap (i.e. E_{g1} in the first quadrant, E_{g2} in the three others). This is illustrated in figure 8.

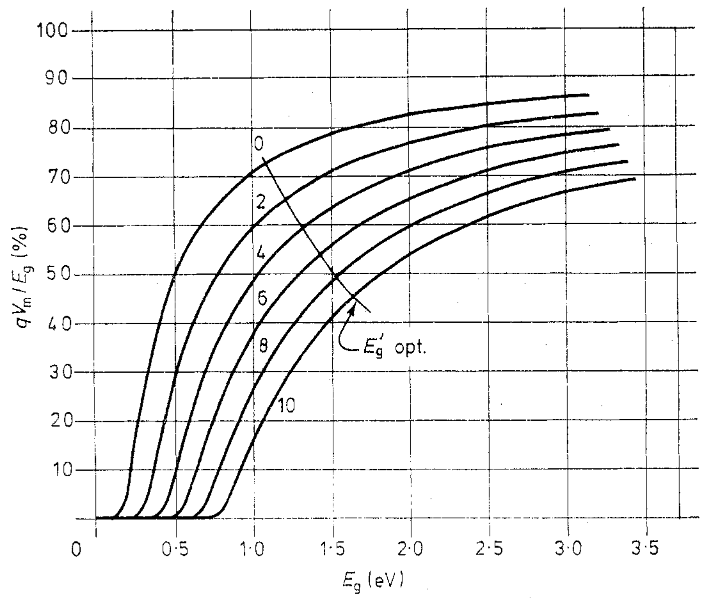

Figure 8. Bias voltage for maximum power V_m for a solar cell with I–V characteristic as defined in figure 1 in function of the band-gap E_g and the a-factor. Curve parameters give values of lg A where A (Wm^{-1})$=ahc$.

5. Application to experimental cells

Our theory has been developed for a p$^+$n heterojunction cell. Various experimental cells are of the n$^+$p type. A completely analogous treatment can be made for this type of cells. In all equations and figures one should change the notations and subscript n by p and c by v. In figure 9 we have summarised the crucial diagram of figure 4. For a p$^+$n cell one should use the (ΔE_c, ΔE_g) diagram; ΔE_v is then represented by the vertical (or

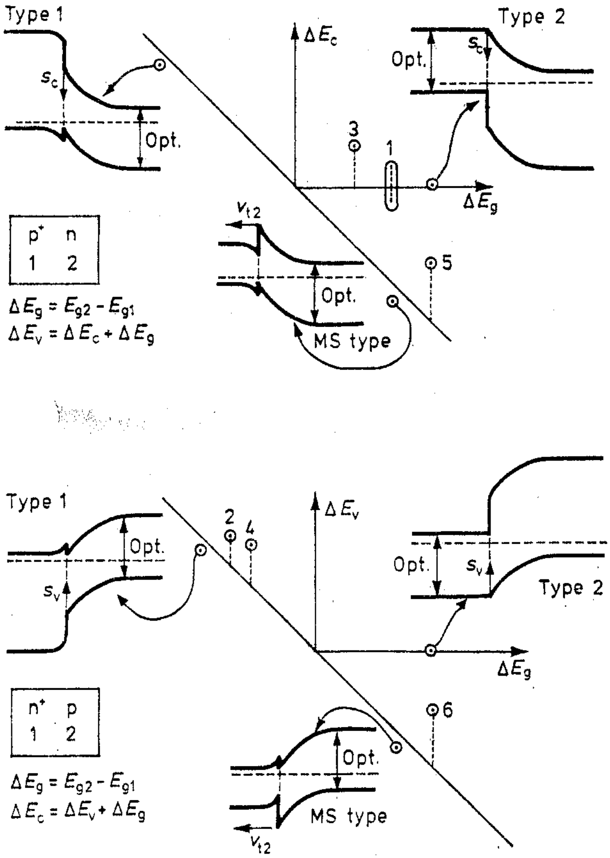

Figure 9. ΔE_c, ΔE_g diagram for p⁺n heterojunctions (ΔE_v is the vertical distance to the bisector) and ΔE_v, ΔE_g diagram for n⁺p heterojunctions (ΔE_c is vertical distance to the bisector). The band-diagrams for the three types of optimum structures are shown with the optimal band-gap and the fundamental process that determines the dark current. The range 1 represents the various models for the p⁺Cu_2S–nCdS cell, the point 2 the n⁺CdS–pInP cell, point 3 the p⁺CdTe–nCdS cell, point 4 the n⁺CdS–pCdTe cell, point 5 the p⁺Ge–nGaAs cell, and point 6 the n⁺Ge–pGaAs cell. Each time the deviation ΔE_c or ΔE_v from optimal conditions is shown in dashed lines.

horizontal) distance to the bisector of the second and fourth quadrant, which shows the onset of inversion; the three types of optimum structures are shown on the diagram; the band-gap that should be chosen optimal and the fundamental process that determines the dark saturation current, and thus the value of the a-factor is indicated in each case. For an n⁺p cell one should use the (ΔE_v, ΔE_g) diagram; ΔE_c is then represented by the vertical distance to the bisector that shows the onset of inversion. Again the three types of optimum structures are shown. It is important to note that if one has a near optimum p⁺n cell of type 2 with a slightly positive ΔE_c, and one then changes the relative dopings to make an n⁺p cell without changing the semiconductors, one obtains a near optimum cell of type 1 with again a slightly positive ΔE_c (i.e. slightly above inversion).

The GaAlAs–GaAs solar cells are described in the literature as heterojunction cells, but in fact they consist of an isotype heterojunction or a graded band-gap semiconductor followed by a homojunction (Sutherland and Hauser 1977). The fundamental solar cell process takes place in the homojunction. The isotype heterojunction or the graded band-gap semiconductor serves to increase the collection efficiency γ at one side of the homojunction by creating a window effect and/or a quasi-electric field. These effects are related to the front or back surface-field effects (Pauwels *et al* 1978, Fossum 1977). In our theory these kinds of effects are included in 'the appropriate choice of geometry' mentioned in the discussion of equation (31).

The p^+Cu_2S–nCdS solar cell is a near optimum structure of the type 2. The band-gap of Cu_2S is near the optimum (close to 1 eV), the band-gap of CdS far above it (2·4 eV). Our theory predicts that these structures depend rather critically on the value of ΔE_c because according to equation (21) the interface transport efficiency δ is close to 1 for $\Delta E_c > 0$ and decreases very rapidly to zero when ΔE_c becomes negative. In fact, in figure 4 we have set $\delta = 1$ for $\Delta E_c > 0$ and $\delta = 0$ for $\Delta E_c < 0$. There is a discrepancy between the published values for ΔE_c. Earlier authors (Lindquist and Bube 1972) claim a negative ΔE_c of the order of 0·5–3 kT. The low efficiency of the cell is explained by the low value of $Q_1 = \delta\gamma$. Tunnelling through the conduction band spike causes a higher value for δ than predicted by our theory. These authors explain the typical quenching and cross-over effects of the Cu_2S–CdS cell by the dependence of this tunnelling on the effective doping of the CdS. As mentioned before, an accurate analysis of the influence of tunnelling on near optimum structures of type 2 will be published elsewhere (De Visschere and Pauwels). Te Velde (1973) suggested a whole range of ΔE_c values, depending on the preparation method, going from strongly positive (with 'bad' values of the a-factor) to strongly negative (with low value of δ). More recent authors (Rothwarf and Barnett 1977, Pavelets and Fedorus 1976) accept a positive value of ΔE_c of about 4 kT. This means that their a-factor is somewhat larger than optimum. They claim however that the drift velocity v_{dr} of electrons at the interface of the CdS is lower than the thermal velocity v_{t2}, in which case one should use the diffusion theory instead of the thermionic emission theory. We will show elsewhere (Pauwels *et al* 1978) that one should then replace δ and s_2 in our theory by $\delta_2\delta$ and $\delta_2 s_2$ where δ_2 is a transport efficiency for electrons through the space charge region of CdS given by

$$\delta_2 = v_{dr}/(v_{dr} + s_2). \tag{42}$$

This means that the a-factor improves somewhat, but the decrease of $Q_1 = \delta_2\delta\gamma$ has a much more drastic effect on the efficiency. Furthermore the value of s_2 is increased by tunnelling through the energy barrier of CdS. These authors explain quenching and cross-over by the dependance of v_{dr} on the effective doping of the CdS. One should notice that this application of our theory is only qualitatively valid because v_{dr} and thus δ_2 depend on the bias voltage such that the I–V characteristic has no longer the accepted exponential V-dependance.

It is clear that the efficiency of these type 2 structures can be improved by lowering s_2 because it improves the a-factor and, according to equation (42), the collection efficiency. It is generally accepted that a better lattice match causes a lower s_2. It is for this reason that one recently has started investigating the $Cu_2S - Cd_{1-x}Zn_xS$ cell (Barnett *et al* 1977, Domens *et al* 1977). It is indeed hoped that $Cd_{1-x}Zn_xS$ should match better to Cu_2S than CdS matches to Cu_2S.

The recently announced n^+CdS–pInP cell seemed very promising because efficiencies of 12·5% have been reached with single crystal InP (Bachmann *et al* 1976, Shay *et al* 1976). Unfortunately its future seems limited because of the scarcity of In. It is a near-optimum n^+–p structure of type 1. The band-gap of InP is near the optimum (1·2 eV). However, the ΔE_c is positive: Shay *et al* give a value of $\Delta E_c = 0{\cdot}56$ eV $= 22\ kT$; our own data based on an electron-affinity of 4·5 eV for CdS and 4·38 eV for InP give $\Delta E_c = 0{\cdot}12$ eV $= 5\ kT$. The authors claim that this positive ΔE_c is beneficial because it does not constitute a barrier for the excited electrons in the InP. Our theory says however, that a negative ΔE_c is better, because it constitutes no barrier for the collection of electrons since they are majority carriers at the interface, and because it gives the a-factor its optimal value; a positive ΔE_c instead increases the a-factor above its optimal value by the factor exp (ΔE_c). The fact that nevertheless a very good efficiency is obtained should be attributed to the excellent lattice match between InP and CdS and thus to a very small s_v. The remark of these authors would be correct if the InP were the heavily doped semiconductor, because then the cell would become a p^+n cell of type 2, and there a strongly positive ΔE_c, although not optimal, is certainly better than an even slightly negative ΔE_c.

Various authors have reported on pCdTe–nCdS cells (Fahrenbruch *et al* 1974, 1977). The CdTe has a near optimum band-gap (1·44 eV). The p^+CdTe–nCdS is of type 2 but with a positive $\Delta E_c = 0{\cdot}22$ eV $= 9\ kT$, so that the a-factor deviates from the optimal value by the factor exp (ΔE_c). The s_c is probably rather large because the lattice match is reported to be 'fair'. Also the n^+CdS–pCdTe cell is described. This cell is then of type 1, and the a-factor deviates by the same factor exp (ΔE_c) from its optimal value.

Various other heterojunction cells are reported in the literature, i.e. pCdTe–$nCd_{1-x}Zn_xS$ (Fahrenbruch *et al* 1974, 1977), nCdS–$pCuInSe_2$ (Shay *et al* 1975, Kazmerski *et al* 1976). The data are however rather qualitative so that it is difficult to situate them on our diagrams. The authors pay attention to a small lattice mismatch (small s_c or s_v) and to the fact that ΔE_c is slightly positive, which means that their cells are of type 2 if they are p^+n cells, and of type 1 if they are n^+p cells. In the second case our theory would again predict that a positive ΔE_c is not needed.

Very few examples of the optimum heterojunction cells of the MS type are mentioned in the literature. In early work (Lopez and Anderson 1964, Donnelly and Milnes 1966) photocurrent experiments are reported on p^+Ge–nGaAs and on n^+Ge–pGaAs, but in the first case ΔE_v and in second case ΔE_c are much too strongly positive for these cells to be optimum. The cells are illuminated from the side of the optimum band-gap material (GaAs). Optimum heterojunction cells of the MS type behave in fact exactly as real MS cells, but real MS cells are always illuminated through the transparent metal.

Heterojunctions in which one of the semiconductors is SnO_2 or In_2O_3 or mixtures also behave more or less as MS cells. These oxides are degenerate n^+type semiconductors with very large band-gaps: they behave as transparent metals. Experiments are reported on SnO_2–GaAs, SnO_2–Ge (Wang and Legge 1976), SnO_2–Si (Perotin *et al* 1976) and $(In_2O_3)_x(SnO_2)_{1-x}$–Si (Burk and Dubow 1976) cells. If the other semiconductor is p type, these cells are n^+–p cells of type 1, but in each case with a much too strongly positive ΔE_c to be optimum. If the other semiconductor is n-type, they are isotype heterojunction cells in which the collection of photon-excited holes occurs by recombination with electrons from the conduction band of the oxides through interface states (figure 6). This type of heterojunctions and this type of collection mechanism is not covered by our theory.

6. Conclusions

Under certain limiting assumptions, listed in the introduction, we have been able to identify optimum heterojunction p^+n or n^+p solar cells. Three types of such structures have the best efficiencies. In the first type, the weakly doped semiconductor should have an optimum band-gap, it should be inverted at the interface, and the strongly doped semiconductor should have a larger band-gap. In the second type, the strongly doped semiconductor must have an optimum band-gap and the discontinuity at the interface in its minority carrier band must approach zero from the positive side, i.e. it may not constitute a barrier for the collection of minority carriers. Many examples of these type of structures are described in the literature, although many of them deviate more or less from the optimum conditions. The third type has the characteristics of the first type but the strongly doped semiconductor has the smaller band-gap. It behaves exactly as MS cells, and we called it MS type cell. Almost no examples of this type were found in the literature. Some heterojunction cells that also behave as MS cells are isotype heterojunction cells and are not covered by our theory.

Appendix

In this appendix the conditions (10*b*) and (10*c*) are proved which express that the drop V_s in E_{Fp} (figure 2) is negligible. This V_s is determined by the continuity equation (6), which reads in terms of V_s.

$$p_1 \exp(-V_1)[1-\exp(-V_s)]v_{t1} \exp(-\Delta E_v) \\ = p_2 v_{d2} [\exp(V) \exp(-V_s) - 1] - \gamma_2 r_2 I_2 - j_{s2}' \tag{A1}$$

for $\Delta E_V > 0$. For $\Delta E_V < 0$ the factor $\exp(-\Delta E_V)$ is missing.

When $\Delta E_V < 0$, the electrons at the interface recombine almost exclusively with the holes of semiconductor 2 so that $j_{s2}' \approx j_{s1} + j_{s2} = j_s$. The terms n_{1s0} and n_{2s0} of equations (12) and (13) are then determined by E_{Fp} at the interface of semiconductor 2, so that our calculations of the electron transport (j_n and j_s) is only correct if $V_s \ll 1$, which then also assures that equation (17) for j_{0p} is correct. Elimination of $\exp(-V_s)$ from (A1) for $\Delta E_V < 0$ leads to

$$\exp(-V_s) = \frac{p_2 v_{d2} + \gamma_2 r_2 I_2 + j_s + v_{t1} p_1 \exp(-V_1)}{p_2 v_{d2} \exp(V) + v_{t1} p_1 \exp(-V_1)}. \tag{A2}$$

The relation between p_1 and p_2 is

$$p_2 \exp(V) = p_1 \exp(-V_1) \exp(-V_2 - \Delta E_V)\, N_{V2}/N_{V1}. \tag{A3}$$

The second condition of (10*b*) and (10*c*) therefore assures that $v_{t1} p_1 \exp(-V_1)$ is the dominant term of the denominator. At normal operation points of a solar cell, $\gamma_2 r_2 I_2$ and j_s (equation 22) can be of the order of the dark current, which is (s_2 or v_{t2}) $\times$ $n_2 \exp(-V_2)$. The first conditions of (10*b*) and (10*c*) therefore assures that $v_{t1} p_1 \exp(-V_1)$ is also the dominant term of the numerator. We conclude that $V_s \ll 1$.

When $\Delta E_V > 0$ the electrons recombine almost exclusively with the holes of semiconductor 1 so that n_{1s0} and n_{2s0} are calculated with E_{Fp} of semiconductor 1. Our calculations for j_n and j_s are therefore correct, but the expression for j_{0p}

$$j_{0p} = p_2 v_{d2} [\exp(V) \exp(-V_s) - 1] - \gamma_2 r_2 I_2 \tag{A4}$$

must be checked. The Shockley–Read theory says that now

$$j_{s2}' \approx \exp(-\Delta E_V) \exp(-V_s)\, j_s. \tag{A5}$$

Elimination of exp $(-V_s)$ from (A1) for $\Delta E_v > 0$ and substitution in (A4) leads to

$$j_{0p} = p_2 v_{d2} [\exp(V) - 1](1/N)\{v_{t1} p_1 \exp(-V_1) - j_s/[\exp(V) - 1]\}$$
$$- \gamma_2 r_2 l_2 (1/N) [v_{t1} p_1 \exp(-V_1) + j_s] \qquad (A6)$$

where

$$N = v_{t1} p_1 \exp(-V_1) + j_s + p_2 v_{d2} \exp(V + \Delta E_v). \qquad (A7)$$

The first conditions of (10*b*) and (10*c*) now assures that equation (17) is correct.

References

Bachmann K J, Buehler E, Shay J L and Wagner S 1976 *Appl. Phys. Lett.* **29** 121

Barnett A M, Meakin J D and Rothwarf A 1977 *Proc. Photovoltaic Solar Energy Conf., Luxembourg* (Dordrecht: Reidel) paper 11.1

Burk D E and Dubow J B 1976 *Proc. 12th IEEE Photovoltaic Specialist Conf., Baton Rouge* (Baton Rouge: IEEE) p 971

De Visschere P and Pauwels H J 1978 *Appl. Phys.* to be published

De Vos A 1976 *Energy Conversion* **16** 67

De Vos A and Pauwels H J 1976 *Proc. Solar Energy Conf., Toulouse* (Toulouse: CNES) p 523

—— 1977 *IEEE Trans. Electron. Dev.* **ED-24** 388

Domens P, Cadene M and Solal G W 1977 *Proc. Photovoltaic Solar Energy Conf., Luxembourg* (Dordrecht: Reidel) paper 11.9

Donnelly J P and Milnes A G 1966 *Int. J. Electron.* **20** 295

Fahrenbruch A L, Aranovich J, Courrages F, Yin S Y and Bube R H 1977 *Proc. Photovoltaic Solar Energy Conf., Luxembourg* (Dordrecht: Reidel) paper 11.8

Fahrenbruch A L, Vasilchenko V, Buch F, Mitchell K and Bube R H 1974 *Appl. Phys. Lett.* **25** 605

Fossum J G 1977 *IEEE Trans. Electron. Dev.* **ED-24** 322

Kazmerski L L, White F R and Morgan G K 1976 *Appl. Phys. Lett.* **29** 268

Lindquist P F and Bube R H 1972 *J. Appl. Phys.* **43** 2839

Loferski J 1956 *J. Appl. Phys.* **27** 777

Lopez A and Anderson R L 1964 *Solid-St. Electron.* **7** 695

Pauwels H J, De Visschere P and Reusens P 1978 *Solid-St. Electron.* to be published

Pavelets S Yu and Fedorus G A 1976 *Sov. Phys.—Semicond.* **10** 1036

Perotin M, Szepessy L, Manifacier J C, Parot P, Fillard J P and Savelli M 1976 *Proc. Solar Energy Conf., Toulouse* (Toulouse: CNES) p 481

Rappaport P 1959 *RCA Rev.* **20** 373

Rothwarf A and Barnett A M 1977 *IEEE Trans. Electron Dev.* **ED-24** 381

Shay J L, Wagner S, Bachmann K J and Buehler E 1976 *J. Appl. Phys.* **47** 614

Shay J L, Wagner S and Kasper H M 1975 *Appl. Phys. Lett.* **27** 89

Sutherland J E and Hauser J R 1977 *IEEE Trans. Electron. Dev.* **ED-24** 363

Sze S 1969 *Physics of Semiconductor Devices* (New York: Wiley) p 378

Te Velde T S 1973 *Solid-St. Electron.* **16** 1305

Wang E Y and Legge R N 1976 *Proc. 12th IEEE Photovoltaic Specialist Conf., Baton Rouge* (Baton Rouge: IEEE) p 967

Wolf M 1960 *Proc. IRE* **48** 1246